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Module 7 Introduction

Lesson 1: Foundations of Bayesian Regression Analysis

✔ **Reading:** Foundations of Bayesian Regression Analysis Readings
4h

✔ **Video:** Foundations of Bayesian Regression Analysis Pt. 1
4 min

✔ **Video:** Foundations of Bayesian Regression Analysis Pt. 2
4 min

✔ **Video:** Foundations of Bayesian Regression Analysis Pt. 3
7 min

✔ **Video:** Foundations of Bayesian Regression Analysis Pt. 4
3 min

✔ **Video:** Foundations of Bayesian Regression Analysis Pt. 5
6 min

✔ **Video:** Foundations of Bayesian Regression Analysis Pt. 6
6 min

✔ **Video:** Foundations of Bayesian Regression Analysis Pt. 7
6 min

✔ **Practice Assignment:** Foundations of Bayesian Regression Analysis Quiz
Submitted

Lesson 2: Advanced Techniques in Hierarchical Linear Models

Lesson 3: Exploring Generalized Linear Models in Bayesian Context

Module 7 Summative Assessment

Module 7 Summary

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

Foundations of Bayesian Regression Analysis Quiz

≡ Instructions

Review Learning Objectives

1.

What assumption is made about the errors in the simplest form of the normal linear model?

1 / 1 point

○ Errors are dependent and have equal variances

● Errors are independent and have equal variances

○ Errors are dependent and have unequal variances

○ Errors are independent and have unequal variances

Submitted

Attempts

Unlimited

○ Retry

✔ **Correct**

Correct! The simplest form of the normal linear model assumes errors are independent and have equal variances.

To pass you need at least 80%. We keep your highest score.

View submission

See feedback

2.

What is the primary purpose of using transformations in linear regression models?

1 / 1 point

● To linearize the relationship between response and explanatory variables

○ To simplify the model-building process

○ To eliminate the need for a prior distribution

○ To increase the number of parameters in the model

Submitted

Attempts

Unlimited

✔ **Correct**

Correct! Transformations are often used to linearize the relationship between the response and explanatory variables

3.

What is the key advantage of using Bayesian methods for linear regression?

1 / 1 point

○ They eliminate the need for model checking

○ They ensure the model is always correctly specified

○ They simplify the computation of the likelihood function

● They allow for incorporating prior knowledge into the model

Submitted

Attempts

Unlimited

✔ **Correct**

Correct! Bayesian methods allow for incorporating prior knowledge into the model.